

Project Development Plan and Milestones

Implementation Timelines and Risk Controls

1. Phased Development Timelines

- **Phase 1: Ideation and Database Setup:** ER diagrams mapping, Mongo schemas writing, database seeding, test dataset verify.
- **Phase 2: Backend REST RESTful APIs:** Controllers logic development, JWT authorization middleware, slot-conflict validator.
- **Phase 3: React Vite Client Components:** Interactive calendar setup, dashboards, global context configs, Axios settings.
- **Phase 4: Integrations and Network Exposure:** Socket.io real-time alerts setup, Vite reverse proxy mapping, local network sharing.

2. Risk Analysis and Risk Resolution Controls

- **Risk: Router Client Isolation:** Friends cannot access IP URLs because router blocks local peer sharing. Solution: Expose Vite via a secure public tunnel (Pinggy) mapping port 5173.
- **Risk: Cookie Session Collision:** Testing multiple dashboards in the same browser pollutes sessions. Solution: Add global Axios interceptor redirecting users to login page upon 401/403 errors.